Dual-Feature Attention-Based Contrastive Prototypical Clustering for Multimodal Remote Sensing Data

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***Abstract*— The integrated use of multisource remote sensing (RS) data in Earth observation missions has garnered considerable attention. Hyperspectral images (HSIs) offer extensive spatial and spectral detail, whereas light detection and ranging (LiDAR) data provide elevation information. Therefore, the fusion of HSI and LiDAR data can enhance the accuracy (ACC) of image classification. However, contemporary supervised multimodal deep learning techniques depend heavily on extensive human-annotated training datasets. To address this challenge, we propose a contrastive prototypical clustering network enhanced with a dual-feature attention module. Specifically, two sets of enhanced modal views are constructed from the multimodal RS images for the subsequent contrastive learning. The proposed dual-feature attention module emphasizes channel and spatial attention separately for each modality, integrating both to adjust the feature representation across different channels and positions. By learning the importance weights of each channel and position, this module highlights the hierarchical structure and enhances the discriminative quality of the features. The learned features are utilized through an online clustering mechanism and a self-supervised training strategy that combines contrastive loss and cluster loss to achieve efficient and effective land cover classification. Extensive experiments on three widely used HSI and LiDAR datasets demonstrate that the proposed method outperforms current state-of-the-art approaches. The code for this method is openly available at: https://github.com/RogsDing/DFCPC.**

***Index Terms*— Attention mechanism, contrastive learning, hyperspectral images (HSIs), light detection and ranging (LiDAR) data, prototypical clustering.**

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I. Introduction

In RECENT years, the remote sensing (RS) imaging technology has advanced significantly, particularly with the development of RS imaging systems, which has greatly enhanced the availability of RS images [1]. Now, not only can we obtain data from multiple RS devices in the same geographical area but we can also utilize the multimodal data to analyze land cover information. For example, hyperspectral image (HSI) sensors provide the detailed spectral information [2], while light detection and ranging (LiDAR) provides the elevation information of ground objects [3]. These sensors capture various attributes of the Earth's surface, such as spectral characteristics and terrain information, thereby furnishing a more comprehensive and accurate dataset for the identification and classification of ground objects. For example, the grass on a roof and the grass on a road may be composed of the same material, making it difficult to distinguish between them using hyperspectral imaging alone. However, the elevation data provided by LiDAR can effectively supplement this information and help differentiate between the two. Wang and Menenti [3] can easily differentiate between the two types of grass by assessing their height relative to the ground. By integrating multimodal data, we can interpret features more accurately than with single-modal data alone. Consequently, the joint application of cross-modal data has garnered significant attention in the RS image interpretation and plays a crucial role in various applications, such as land cover classification [4], target detection [5], [6], [7], and environmental monitoring [8].

With the rapid development of deep learning technology, deep neural network models have been widely adopted in the field of RS image processing. For the joint classification of HSI and LiDAR data, researchers have proposed various deep learning models [9], [10], [11]. In addition to traditional feature-level fusion methods, decision-level fusion has also emerged as a popular approach for HSI and LiDAR classification tasks. These methods derive final classification results by employing multiple classifiers and utilizing voting strategies. In recent years, there has been an increasing interest in integrating deep learning technology with multisource data for joint classification. Techniques, such as encoder–decoder networks [12], [13], coupled convolutional neural networks (CNNs) [14], Gabor CNNs [15], cross-attention mechanisms [16], and transformers [17], [18] have been explored. These deep learning models have shown signifi-

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cant success in joint classification tasks involving multisource data and have found widespread practical applications [19], [20], [21], [22].

The deep learning has made notable strides in the joint classification of multisource data; however, it still faces challenges in developing effective HSI and LiDAR classification models. Traditional supervised learning methods depend on large volumes of labeled samples to maintain accuracy (ACC), which is costly and limits feature modeling capabilities. Recently, self-supervised learning has emerged as a promising approach to address this issue by extracting intrinsic attributes and semantics from large-scale unlabeled data, thereby reducing the need for costly labeled data [23], [24]. In joint HSI and LiDAR classification, self-supervised learning has garnered attention as a potential solution, though research in this area remains limited.

Effective multimodal clustering depends on integrating information from multiple modalities. Early fusion techniques, such as crosstalk and weighted sums, often oversimplified intermodal interactions and neglected complex relationships. In contrast, attentional mechanisms [25] have demonstrated remarkable success in various visual-language tasks [26]. Among them, Transformer [27] exemplifies the successful application of the attention mechanism, offering strong generalization capabilities and serving as a crucial component in state-of-the-art multimodal learning models. However, Transformers often require substantial computational resources for training, making them computationally expensive. In addition, while they primarily process global information through the self-attention mechanism, this can sometimes result in relatively insufficient handling of local information. Traditional multiview and multisensor clustering methods, while effective, often rely on shallow or handcrafted features and use offline clustering mechanisms.

The online prototype clustering is designed to handle continuously arriving data streams, in contrast to offline clustering, which operates on static datasets. This approach enables the model to adapt to new data points promptly. By maintaining a set of representative data points or prototypes, each one serves as the center for a cluster. New data points are assigned to the nearest prototype, thereby forming clusters. This method reduces storage requirements, as only a limited number of prototypes need to be retained instead of the entire dataset. As new data points arrive, the clustering model updates the prototypes incrementally. This updating typically involves repositioning existing prototypes or adding new ones to ensure that the clustering structure accurately reflects the most recent data distribution.

Our model references Transformer-based contrastive prototypical clustering for multimodal RS data (TMPCC) [28], which designs a multimodal transformer to learn a shared space through adaptive intermodal and intramodal interactions. However, the transformer model is more complex, with higher computational overhead and more parameters. When processed modal features have significant channel and spatial importance, combining attentional mechanisms can more accurately capture both local and global information, enhancing the feature representation effectiveness. While transformers excel at modeling long-range dependencies, for certain tasks and datasets, the local information and simplified structure offered by channel and spatial attention mechanisms may be sufficient and more efficient. By employing a simplified attention mechanism, the number of model parameters can be reduced, which mitigates the risk of overfitting and enhances training efficiency.

To address these challenges, we introduce a new clustering framework: the dual-feature attention module-based contrastive prototypical clustering (DFCPC) network. This framework emphasizes intermodal interactions, enhances the processing of both local and global information and highlights the hierarchical structure while increasing feature discriminability. Second, we use contrastive learning to enhance representations through extended views generated by data augmentation, which helps the model capture richer semantic information and features, thereby improving performance in complex tasks. Finally, our prototype-based online clustering network dynamically adjusts clustering prototypes based on data characteristics and changes, offering an effective solution for real-time data streams and dynamic environments.

The main contributions of this article are summarized as follows.

1) We introduce DFCPC, a clustering framework for multimodal RS data that incorporates prototype learning, contrastive learning, and a dual-feature attention module. This framework enhances the ACC of intermodal semantic alignment by emphasizing the localization of different features through channel attention and spatial attention.
2) We propose a dual-feature attention module that focuses on both the channel and spatial dimensions of the feature map. By concatenating channel and spatial attention, this module comprehensively considers feature information and enhances feature representation capability. As the channel and spatial attention mechanisms operate in parallel, the feature map retains more information during processing, minimizing the information loss associated with conversion steps.
3) The interactive integration of channel attention and spatial attention modules enables the model to learn the global interdependence between channel and spatial features. It adjusts the feature representation of various channels and locations by learning the importance weights of each, which underscores the hierarchical nature of the structure and enhances the discriminative power of the features.
4) We conducted extensive experiments on three benchmark datasets, showing its superiority over existing methods.

II. Related Works

A. Contrastive Learning

In recent years, the contrastive learning has become a key technique in RS image processing and analysis [29], [30]. This method involves comparing multiple augmented views of

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the same sample, treating them as similar pairs, while views from different samples are treated as dissimilar pairs. The contrastive loss function ensures that different transformations of the same sample are as close as possible in feature space, while transformations from different samples are as distant as possible, thereby learning invariant and discriminative feature representations [31]. Contrastive learning is widely used for unsupervised representation learning in images, as demonstrated by MoCo [32] and SimCLR [33], which have shown significant improvements on the ImageNet dataset.

In the hyperspectral data processing, researchers have employed various enhancement methods to construct views and utilize contrastive learning for feature extraction during pretraining, such as for land cover classification [34]. Besides contrastive learning within the spatial domain, there are cross-domain contrastive learning frameworks designed to learn domain-invariant features [35]. These methods have shown excellent performance and generalization capabilities in various tasks, including RS image scene classification [36], semantic segmentation [37], polarimetric synthetic aperture radar (PolSAR) land cover classification [38], synthetic aperture radar (SAR)-optical data fusion [39], and change detection [40].

The self-supervised learning not only reduces the reliance on manually labeled data but also offers an effective approach for training deep clustering models. Contrastive clustering [41], a pioneer in self-supervised learning, has demonstrated excellent performance in unimodal clustering tasks. In the field of RS, the self-supervised learning has made significant strides, particularly in unmixing and classification tasks, and shows promise in bridging the gap between unsupervised and supervised learning.

B. Multimodal Clustering

The multimodal clustering involves grouping and analyzing data from various modalities, such as images, text, and audio. This task typically requires efficiently clustering data samples across different modalities for further analysis and applications [42], [43], [44]. Self-supervised learning methods can extract feature representations with semantic consistency from multimodal data. These representations aid in identifying similarities and differences between data from various modalities during the clustering process.

Multiview subspace clustering (MVSC) [45] is an effective approach for handling multimodal data. It assumes that data from different modalities can be represented in a low-dimensional common subspace or have consistent representations. A key challenge of MVSC is its computational complexity, particularly the high memory and time requirements when processing large-scale data. To address the computational challenges of MVSC methods on large-scale data, several scalable techniques have emerged. These techniques often use anchor graph methods to approximate the subspace structure, reducing both computational complexity and memory requirements. Large-scale MVSC (LMVSC) [46] addresses the challenges of large-scale datasets by optimizing consistent anchors to enhance clustering efficiency and ACC. Efficient one-pass multiview subspace clustering with consensus anchors (EOMSC-CAs) [45] uses consistency anchors to enhance computational efficiency, making it well-suited for large-scale data. Fast parameter-free multiview subspace clustering with consensus anchor guidance (FPMVS-CAG) [47] also employs consistency anchors but requires no prior parameterization. Scalable multiview subspace clustering (SMVSC) with unified anchors [48] introduce uniform anchor points to ensure consistency across views. Although these methods have made significant strides in managing large-scale data and enhancing computational efficiency, they predominantly address linear and shallow feature representations. This focus limits their effectiveness when handling complex and nonlinear multimodal data.

Deep canonical correlation analysis (DCCA) [49] seeks to learn correlations between multiple datasets by maximizing the correlation between the outputs of several neural networks. This approach enables the capture of more complex relationships between datasets at a deep level. Autoencoder (AE) [50], [51], [52], [53] is designed to learn a compressed representation of data through its encoder and decoder components, often used for feature extraction and data reconstruction. Traditional approaches, such as AE networks (AE2-Nets) [54] and multisensor deep clustering (MDC) [55], typically involve two phases: intra- or cross-modal feature learning in the first phase, followed by clustering or other tasks in the second phase. This approach can result in suboptimal feature representations due to the separate execution of clustering and feature learning. To address these limitations, our proposed method integrates adaptive modal fusion, discriminative representation learning, and online clustering into a unified model.

III. Methodology

In this section, we describe the proposed DFCPC network. We first apply predefined data augmentation to generate two distorted views for each modality. These distorted views are then fed into the dual-feature attention module, with the outputs subsequently passed to the prototype-based clustering head for categorization. The overall framework of DFCPC is illustrated in Fig. 1, and further details will be provided in Sections III-A–III-D.

A. Data Augmentation

To train our DFCPC, we employ several simple yet effective enhancement strategies tailored to the spatial and spectral features of RS images. These strategies include random cropping, masking, resizing, flipping, and Gaussian blurring. These methods facilitate the model's ability to learn features at different locations and scales, thereby enhancing its capacity to capture both local details and the overall image structure. We apply random band selection and random band substitution to multichannel spectral images to ensure consistency and optimize model performance across various data sources. In addition, by randomly combining these transformations, we enhance the diversity of the training data and improve the model's robustness. This approach enables the model to better capture underlying patterns and semantic information, thus improving its adaptability and generalization in complex environments.

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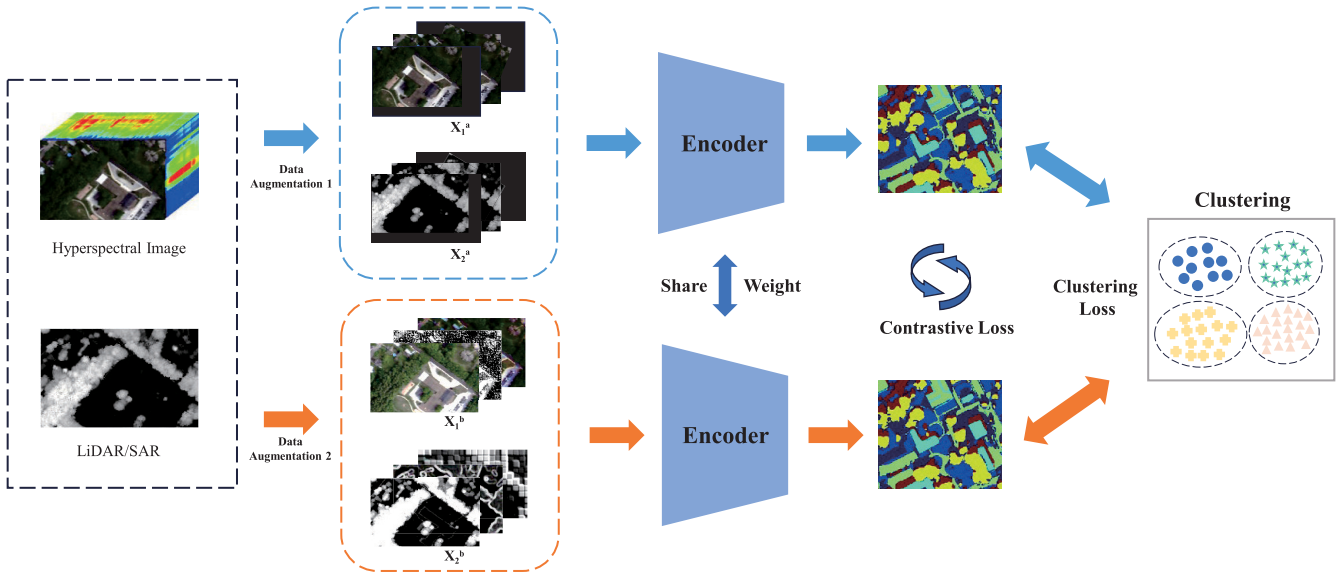


Fig. 1. Overview of the TMPCC framework. It consists of three components. (a) Data augmentations. (b) Shared twin backbone network (c) Training with a combined contrastive and clustering loss.

Algorithm 1 PyTorch-Style Pseudocode for DFCPC

```
# h: attention encoder
# g: clustering head
# loss_ins: instance contrastive loss
# loss_clu: clustering loss
# tau, lambda: loss hyper-params
# N: batch size; 7 × 7: patch size
# training phase
# load a batch with N samples M modalities
for m1, m2 in loader: # e.g., 2 modalities
   # random augmentations of m1 and m2
   m1_a, m1_b = augment(m1) # N × 7 × 7 × D1
   m2_a, m2_b = augment(m2) # N × 7 × 7 × D2
   # forward data
   y_a = g(h(m1_a, m2_a)) # N × C
   y_b = g(h(m1_b, m2_b)) # N × C
   # loss
   l1 = loss_ins(y_a,y_b,tau)
   l2 = loss_clu(y_a,y_b)
   loss = l1 + lambda*l2
   # optimization step
   loss.backward() # back-propagate
   optimizer.step() # Adam update
# inference phase
pred = g(h(m1, m2))
labels = argmax(pred, dim = 1)
```

B. Dual-Feature Attention Module

Recent advancements in attention mechanisms have significantly enhanced deep neural networks, making them widely applicable to cross-modal tasks, such as visual question answering, image description, and graph matching. This success has stimulated interest in recent research approaches that utilize attention mechanisms to facilitate interaction and information exchange between multimodal RS data [56], [57], [58], [59]. A typical attention module, convolutional block attention module (CBAM) [60], enhances feature extraction for CNNs by combining channel attention and spatial attention. However, CBAM performs this fusion in a simple tandem manner, analyzing channel and spatial attention in isolation and not considering their interdependence.

For this reason, we propose a dual-feature attention module that jointly models channel and spatial attention, enabling them to interact directly and explore their potential dependencies. As shown in Fig. 2, the overall structure of the dual-feature attention module is illustrated.

Given the hyperspectral feature maps $\mathbf{F_H} \in R^{C\times H\times W}$ and $\mathbf{F_L} \in R^{C\times H\times W}$ as inputs, they are sequentially processed through two 3×3 convolutional layers $\varphi^1_{3\times3}$ and $\varphi^2_{3\times3}$ for feature extraction. The outputs from these layers are mapped into the same feature space and combined to obtain preliminary fusion results. The fused features $\mathbf{F_{fuse}}$ are then used to construct different scale feature inputs $\mathbf{F_c} \in R^{C\times H\times W}$ and $\mathbf{F_s} \in R^{C\times H\times W}$. The feature map $\mathbf{F_c}$ is input into the channel attention module, while $\mathbf{F_s}$ is input into the spatial attention module

$$\mathbf{F_{fuse}} = \mathrm{Concat}\left(\varphi^1_{3\times3}(\mathbf{F_H}), \varphi^2_{3\times3}(\mathbf{F_L})\right). \tag{1}$$

1) Channel Attention Module: We first aggregate the spatial information of the feature maps using average pooling and maximum pooling operations to generate two different spatial context descriptors: $\mathbf{F_c^{avg}}$ and $\mathbf{F_c^{max}}$, which denote the average pooled features and maximum pooled features, respectively. These descriptors are then passed separately through a multilayer perceptron (MLP) separately, and finally, the output channel feature vectors $\mathbf{F'_c} \in R^{C\times1\times1}$ are merged using principal element summation

$$\mathbf{F'_c} = \mathrm{MLP}(\mathrm{AvgPool}(\mathbf{F_c})) + \mathrm{MLP}(\mathrm{MaxPool}(\mathbf{F_c})). \tag{2}$$

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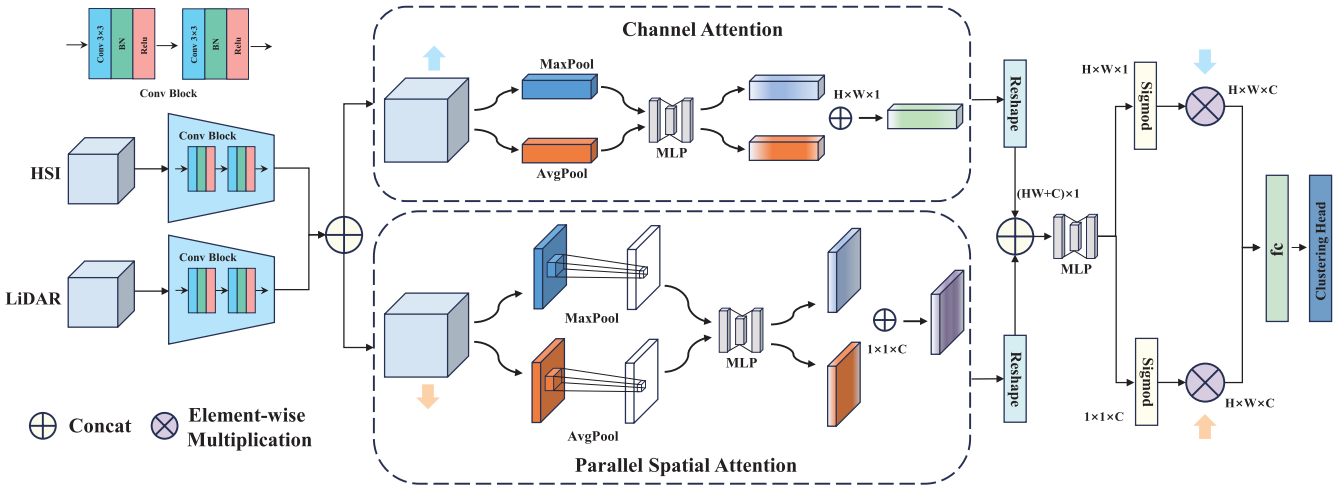


Fig. 2. Overall structure of the dual feature attention module.

2) Spatial Attention Module: We also use average pooling and maximum pooling operations to aggregate the channel information of the feature graph, generating two different channel context descriptors: $\mathbf{F_s^{avg}}$ and $\mathbf{F_s^{max}}$, which represent the average-pooled features and the maximum-pooled features, respectively. The two descriptors are then forwarded to the MLP separately, and finally, the output spatial feature vectors $\mathbf{F'_s} \in R^{1\times H\times W}$ are merged using principal element summation

$$\mathbf{F'_s} = \mathrm{MLP}(\mathrm{AvgPool}(\mathbf{F_s})) + \mathrm{MLP}(\mathrm{MaxPool}(\mathbf{F_s})). \quad (3)$$

3) Interaction of Attention Modules: The obtained channel feature vector $\mathbf{F'_c}$ and spatial feature vector $\mathbf{F'_s}$ are first dimensionally transformed and then concatenated into the MLP network to model the correlation between the channel descriptors and spatial descriptors

$$\mathbf{F'_{fuse}} = \mathrm{MLP}\left(\mathrm{Concat}\left(\mathrm{Reshape}\left(\mathbf{F'_c}\right), \mathrm{Reshape}\left(\mathbf{F'_s}\right)\right)\right) \quad (4)$$

where $\mathbf{F'_{fuse}} \in R^{1\times(HW+C)}$ denotes the fusion feature obtained after interaction.

Next, the interacted features are partitioned into two components $\mathbf{F_{fc}} \in R^{C\times1\times1}$ and $\mathbf{F_{fs}} \in R^{1\times H\times W}$. The original dimensions are restored, and weights are computed using a sigmoid function. These weights are then used for element-wise multiplication with the corresponding inputs. Finally, the resulting outputs from the two parts are combined to produce the final output

$$\mathbf{F_{out}} = (\sigma(\mathbf{F_{fc}}) \otimes \mathbf{F_c}) + (\sigma(\mathbf{F_{fs}}) \otimes \mathbf{F_s}) \quad (5)$$

where σ is the sigmoid activation, $\otimes$ denotes the element-by-element multiplication, and $\mathbf{F_{out}} \in R^{C\times H\times W}$ denotes the final fusion result.

C. Objective Function

This section introduces a loss function for DFCPC during the training phase of self-supervised clustering networks. The objective function is $L = L_{\mathrm{ins}} + \lambda L_{\mathrm{clu}}$, where L_{ins} and L_{clu} denote the instance-level comparison loss and clustering loss, respectively. We use the standard InfoNCE [56] loss for L_{ins}, where L_{clu} is defined as the Kullback–Leibler (KL) scatter between the predicted probability and the auxiliary probability.

1) Instance Contrastive Loss: For a distorted view sample $\mathbf{y_j^a}$ and $\mathbf{y_j^b}$ in a small batch of data, we will call the positive sample pair $(\mathbf{y_j^a}, \mathbf{y_j^b})$ anchored by the j_ath sample. In addition to this, there are $(2N-2)$ pairs of instances considered as negative sample pairs relative to this positive sample pair. Given $\mathrm{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{uv^T}/||\mathbf{u}||||\mathbf{v}||$ as the cosine similarity measure between two feature representations, the instance contrastive loss associated with $(\mathbf{y_j^a}, \mathbf{y_j^b})$ is computed as follows:

$$\mathbf{L_{ins}^{j_a}} = \frac{\exp\left(\mathrm{sim}\left(\mathbf{y_j^a}, \mathbf{y_j^b}\right)/\tau\right)}{\sum_k^{2N} l_{k\neq j}\exp\left(\mathbf{sim}\left(\mathbf{y_j^a}, \mathbf{y_k}\right)/\tau\right)} \quad (6)$$

where τ is the temperature parameter and $l_{k\neq j} \in \{0, 1\}$ is an indicator function with a value of 1 when $k \neq j$. The complete L_{ins} is computed across all positive pairs centered by j_a and j_b, i.e.,

$$L_{\mathrm{ins}} = \frac{1}{2N}\left(L_{\mathrm{ins}}^{j_a} + L_{\mathrm{ins}}^{j_b}\right). \quad (7)$$

The instance comparison loss is designed to facilitate the formation of discriminative feature spaces by minimizing the similarity distance of positive instance pairs and simultaneously maximizing the similarity distance of negative instance pairs.

2) Clustering Loss: We first define an auxiliary probability matrix $\mathbf{P}$ based on the predicted soft assignment matrix $\mathbf{Y}$. Each element of $\mathbf{P}$ is computed as follows:

$$\mathbf{P}_{\mathrm{jc}} = \frac{\mathbf{y}_{\mathrm{jc}}^2/\rho_c}{\sum_{c'}\mathbf{y}_{\mathrm{jc'}}^2/\rho_{c'}} \quad (8)$$

where $\rho_c = \sum_j \mathbf{y}_{\mathrm{jc}}^2$ serves as the soft cluster frequencies approximated within the mini-batch. This auxiliary probability sharpens the soft-assignment probability $\mathbf{y}_{\mathrm{jc}}$ by squaring it and then normalizes it by the associated cluster frequency.

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With this auxiliary probability, we define the clustering loss by minimizing the KL divergence between $\mathbf{P}$ and $\mathbf{Y}$, i.e.,

$$\mathrm{KL}(\mathbf{P} \parallel \mathbf{Y}) = \sum_{j}^{2N} \sum_{c}^{C} p_{\mathrm{jc}} \log \frac{p_{\mathrm{jc}}}{y_{\mathrm{jc}}}. \tag{9}$$

Both ($\mathbf{P^a}$, $\mathbf{P^b}$) and ($\mathbf{Y^a}$, $\mathbf{Y^b}$) are stacked into single matrices, i.e., $\mathbf{P}$ and $\mathbf{Y}$, with a size of $2N$. The goal of the clustering loss is to encourage high-confidence label assignments.

D. Clustering Head and Inference of Clustering

We introduce a prototype-based online clustering header, $\mathbf{Y} = g_\mu(\mathbf{Z})$, to predict soft label assignments for input samples based on shared feature representations. This method is particularly suited for multimodal RS data, where prototypes can encapsulate high-level semantic information that is common across different modalities. For instance, a prototype might integrate both spectral and geometric information to infer land cover types. In addition, these prototypes are generalizable and portable. Specifically, by parameterizing the C learnable prototypes in the shared feature space $\mu \in R^{C \times D_z}$, we adopt the strategy proposed in [50] and [51], which utilizes student's t-distribution to measure the probability distribution of clustered labels. The soft label y_{jc} of the jth sample assigned to the cth cluster is defined as follows:

$$\mathbf{y}_{\mathrm{jc}} = g_\mu(\mathbf{Z_j}) = \frac{\left(1 + \|\mathbf{z_j} - \mu_{\mathbf{c}}\|_2^2/\alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{c'}^{C}\left(1 + \|\mathbf{z_j} - \mu_{\mathbf{c'}}\|_2^2/\alpha\right)^{-\frac{\alpha+1}{2}}} \tag{10}$$

where α represents the student's t-distribution's degree of freedom. In this work, we fix $\alpha = 1$ for all experiments.

The learned soft assignment can be used to infer clustering labels by grouping a sample to the closest prototype. We denote the predicted label as follows:

$$\mathbf{c_j} = \underset{c}{\mathrm{argmax}}\left(y_{\mathrm{jc}}\right), c = 1, \ldots, C. \tag{11}$$

IV. Experimental Results and Analysis

A. Datasets and Evaluation Metric

1) HS-LiDAR Trento Dataset: This dataset was captured over a rural area south of the city of Trento, Italy, by the AISA Eagle sensor. It contains HS and LiDAR data with a spatial sampling of 1 m and a size of 600 × 166 pixels. The HS data consists of 63 spectral bands ranging between 0.40 and 0.98 μm, while the LiDAR data contains two channels. There are six different land cover types in this scene. Table I lists the land cover types for the HS-LiDAR Trento dataset.

2) HS-LiDAR MUUFL Dataset: The dataset was collected in November 2010 at the University of Southern Mississippi's Gulfport campus in Long Beach, MS, USA. The dataset contains on-campus HSI-based and LiDAR-based DSMs with a size of 325 × 220 pixels. The HSI was collected by the ITRES Research Ltd. (ITRES) Compact Airborne Spectral Imager (CASI-1500) sensor with 64 available spectral channels ranging from 0.38 to 1.05 μm with a spatial resolution of 0.54 × 1.0 m. LiDAR data are captured by an ALTM sensor using a laser at 1064-nm wavelength. The spatial resolution is 0.60 × 0.78 m. Table II lists the land cover types for the HS-LiDAR MUUFL dataset.

TABLE I
Distribution of Samples for the Trento Dataset

No.	Class Name	Samples
1	Apple trees	4034
2	Buildings	2903
3	Ground	479
4	Woods	9123
5	Vineyard	10501
6	Roads	3174
	Total	30214

TABLE II
Distribution of Samples for the MUUFL Dataset

No.	Class Name	Samples
1	Trees	23246
2	Mostly grass	4270
3	Mixed ground surface	6882
4	Dirt and sand	1826
5	Road	6687
6	Water	466
7	Building shadow	2233
8	Building	6240
9	Sidewalk	1385
10	Yellow curb	183
11	Cloth panels	269
	Total	53687

TABLE III
Distribution of Samples for the Berlin Dataset

No.	Class Name	Samples
1	Forest	54954
2	Residential Area	268642
3	Industrial Area	19566
4	Low Plants	59282
5	Soil	17426
6	Allotment	13305
7	Commercial Area	24824
8	Water	6672
	Total	464671

3) HS-SAR Berlin Dataset: The simulated EnMAP data synthesized based on the HyMapHS data graphically describes the Berlin urban and its rural neighboring area at 30-m GSD, which can be freely downloaded from the website2. In detail, there are 797 × 220 pixels in this scene, where 244 spectral bands are given in the wavelength range of 0.4–2.5 μm. Table III lists the land cover types for the HS-SAR Berlin dataset.

To quantitatively assess clustering performance, we use five widely accepted metrics: overall ACC, Kappa coefficient (Kappa), normalized mutual information (NMI), adjusted rand index (ARI), and purity. OA denotes the ratio of the model's correct predictions to the overall number on all test samples. Kappa is the percentage of agreement corrected by the number of agreements that would be expected purely by chance. NMI is used to evaluate the agreement between two clusterings or classifications by measuring the amount of information shared between them. ARI is used to evaluate the similarity between two data clusterings. Purity is a metric used to evaluate clustering performance by measuring the extent to which clusters contain members of a single class. Higher scores on these metrics indicate better clustering results. To evaluate

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TABLE IV
CLASSIFICATION ACC (%) ON THE TRENTO DATASET. BEST RESULTS ARE MARKED IN BOLD

Class	LMVSC	EOMSC-CA	BMVC	FPMVS-CAG	SMVSC	DCCA	AE2-Nets	MDC	TMPCC	Ours
Apple trees	74.10	4.02	74.12	25.47	40.78	61.35	53.33	80.34	98.81	**98.74**
Buildings	91.37	0.45	0.23	1.23	2.05	60.87	36.36	99.93	**100.0**	**100.0**
Ground	**89.07**	1.33	5.87	8.45	15.63	0.12	35.48	5.01	3.97	3.76
Woods	97.32	99.35	97.89	98.12	95.42	99.10	20.41	98.99	**100.0**	**100.0**
Vineyard	59.32	68.92	45.12	72.78	70.89	71.52	35.48	60.73	98.69	**98.86**
ACC	77.41	63.17	64.23	68.12	67.65	73.52	40.89	80.31	92.22	**94.42**
Kappa	71.45	49.95	55.07	57.34	57.93	64.99	34.90	75.18	89.64	**92.54**
NMI	67.8	50.37	60.78	53.45	56.78	63.24	22.40	76.98	89.84	**91.20**
ARI	64.94	48.82	55.38	53.67	58.12	62.18	49.78	73.30	93.62	**94.41**
Purity	81.82	69.41	75.34	72.23	74.21	77.46	28.67	91.71	94.18	**94.48**
Running Time(s)	124.97	388.55	156.04	1304.91	86.64	890.47	183.52	31.91	1715.74	512.83

TABLE V
CLASSIFICATION ACC (%) ON THE MUUFL DATASET. BEST RESULTS ARE MARKED IN BOLD

Class	LMVSC	EOMSC-CA	BMVC	FPMVS-CAG	SMVSC	DCCA	AE2-Nets	MDC	TMPCC	Ours
Trees	80.35	0.10	86.75	0.00	**87.60**	54.31	33.33	41.03	60.33	79.83
Mostly grass	**53.58**	0.00	20.36	0.00	0.00	4.09	2.86	45.42	4.54	0.00
Mixed ground surface	59.79	91.67	**93.77**	89.05	1.61	77.77	16.00	60.84	86.41	0.10
Dirt and sand	11.12	**92.97**	8.55	92.38	89.22	0.00	47.62	0.00	0.00	0.00
Road	45.72	0.00	23.22	0.00	83.81	84.55	4.35	85.20	**93.94**	84.73
Water	22.15	**90.83**	10.24	2.45	7.05	0.96	0.00	0.00	1.07	0.00
Building shadow	41.56	4.14	32.39	5.58	4.05	58.52	11.11	25.39	65.02	**81.82**
Building	12.29	0.00	29.55	3.56	4.55	1.11	0.00	90.82	1.23	**96.44**
Sidewalk	12.18	2.47	**54.45**	2.26	3.33	0.91	41.67	0.00	1.01	0.29
Yellow curb	45.16	54.78	0.00	**93.05**	0.43	0.00	5.00	0.00	0.00	0.00
Cloth panels	0.00	**93.83**	11.15	58.47	66.36	0.00	4.35	0.00	0.00	0.74
ACC	46.25	28.16	40.37	29.04	29.26	46.94	22.85	45.31	52.15	**59.76**
Kappa	38.73	23.29	31.35	23.85	23.44	36.98	4.55	34.89	41.09	**48.42**
NMI	49.79	45.59	49.47	42.57	41.84	46.68	7.43	38.66	**51.87**	51.34
ARI	28.68	19.81	28.85	19.59	19.87	36.52	9.12	23.10	40.58	**50.26**
Purity	49.28	26.52	44.79	30.53	30.24	58.84	14.28	54.68	**65.38**	65.03
Running Time(s)	119.65	704.64	950.25	1256.51	1004.64	401.25	302.93	24.80	1223.62	881.90

TABLE VI
CLASSIFICATION ACC (%) ON THE BERLIN DATASET. BEST RESULTS ARE MARKED IN BOLD

Class	LMVSC	EOMSC-CA	BMVC	FPMVS-CAG	SMVSC	DCCA	AE2-Nets	MDC	TMPCC	Ours
Forest	55.60	**90.65**	69.10	24.01	5.12	82.05	22.27	77.45	85.37	86.37
Residential Area	33.25	44.62	43.88	35.46	40.56	51.91	11.48	41.41	**57.11**	54.85
Industrial Area	**49.40**	19.50	0.00	39.37	36.95	0.04	38.29	35.29	0.04	0.05
Low Plants	29.15	32.05	62.46	55.08	49.84	74.18	23.53	51.45	52.14	**78.08**
Soil	50.11	0.00	79.27	80.56	78.72	94.14	12.17	0.00	0.00	**99.09**
Allotment	**53.63**	1.90	0.01	9.50	18.50	0.01	23.16	0.00	0.03	0.01
Commercial Area	5.15	3.96	63.49	1.06	0.85	75.47	22.69	0.69	**81.40**	79.37
Water	17.05	29.94	0.02	1.27	0.43	0.02	40.69	**49.94**	36.05	0.02
ACC	35.94	45.13	47.88	39.15	34.21	56.85	29.22	41.91	54.63	**59.85**
Kappa	24.93	28.57	37.02	15.12	12.51	43.95	21.15	24.31	37.76	**46.27**
NMI	26.75	29.96	36.51	16.93	21.03	43.31	17.64	25.92	41.43	**45.64**
ARI	16.11	22.78	28.98	10.35	11.54	34.42	26.10	17.14	34.19	**36.23**
Purity	63.34	59.26	61.20	50.82	50.87	72.68	30.22	63.23	71.97	**76.50**
Running Time(s)	227.184	526.632	9505.454	2417.362	1231.874	287.726	391.8384	324.2217	14564.117	11881.817

clustering outcomes, we follow a standard procedure by first using the Hungarian algorithm [47] to align the predicted labels with the ground truth labels before calculating the metrics.

B. Implementation Details

For training, we extract 7×7 spatial blocks from each modality to get our training samples. Throughout the training process, we use the optimizer. The model was trained for up to 100 epochs, using 256 batches with a weight decay parameter of 5×10^{-4}. We reduced the learning rate linearly from 5×10^{-4} by a scale factor of 0.1 per 10 epochs. The hyperparameters $\lambda = 1$ and $\tau = 0.5$ were kept constant in all training experiments. The implementation uses PyTorch and runs on an NVIDIA RTX 3090 equipped with 24 GB of graphics memory.

LMVSC[1] is a MVSC method designed to learn subspaces from various viewpoints or modalities, enhancing clustering ACC. It improves upon traditional clustering algorithms by jointly learning subspaces from high-dimensional data across multiple views.

[1]https://github.com/sckangz/LMVSC

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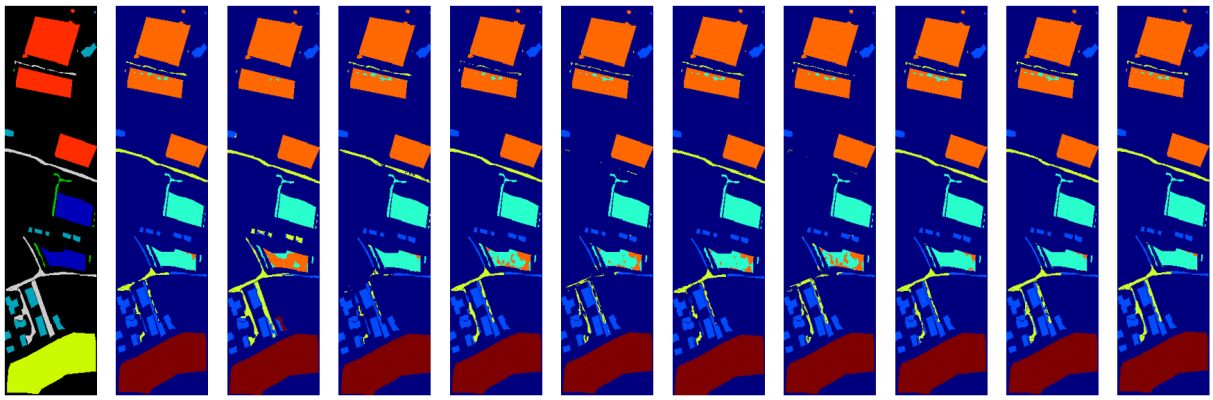


Fig. 3. Clustering maps for the Trento dataset obtained by different clustering approaches. (a) GT. (b) LMVSC. (c) EOMSC. (d) BMVC. (e) FPMVS. (f) SMVSC. (g) DCCA. (h) AE2Nets. (i) MDC. (j) TMPCC. (k) Ours.

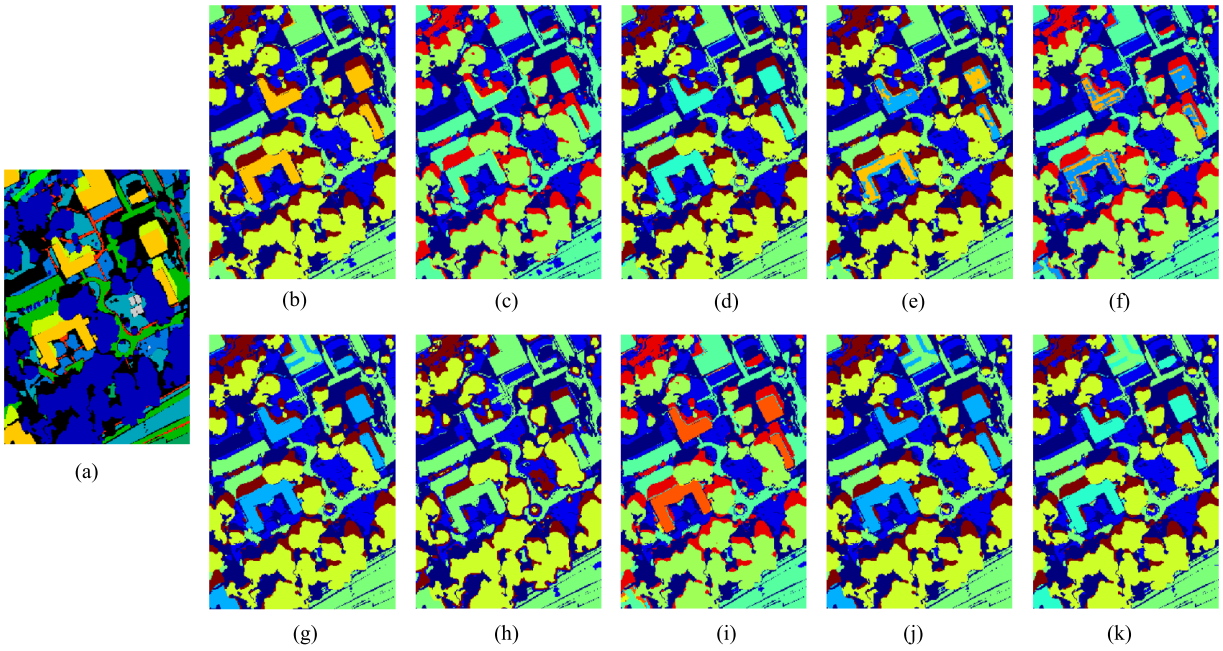


Fig. 4. Clustering maps for the MUUFL dataset obtained by different clustering approaches. (a) GT. (b) LMVSC. (c) EOMSC. (d) BMVC. (e) FPMVS. (f) SMVSC. (g) DCCA. (h) AE2Nets. (i) MDC. (j) TMPCC. (k) Ours.

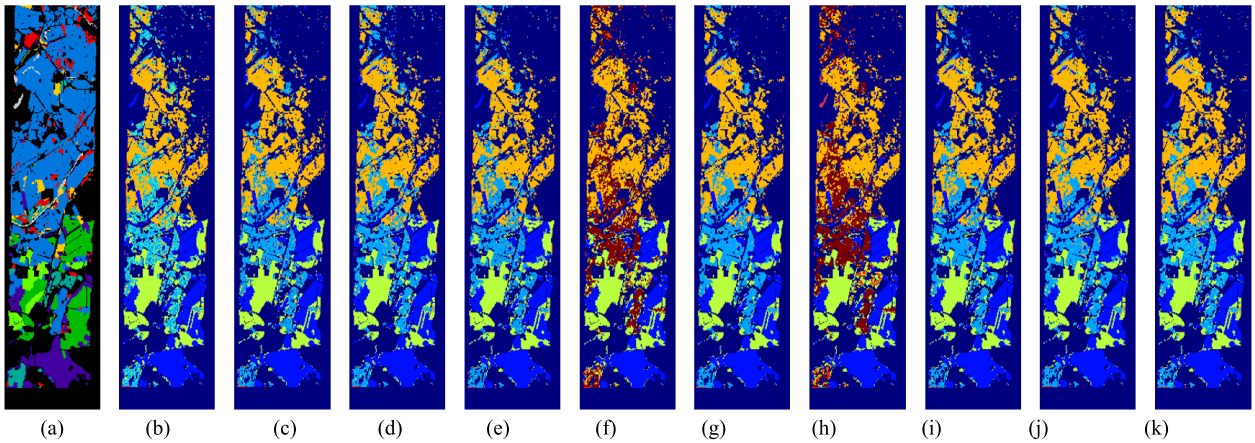


Fig. 5. Clustering maps for the Berlin dataset obtained by different clustering approaches. (a) GT. (b) LMVSC. (c) EOMSC. (d) BMVC. (e) FPMVS. (f) SMVSC. (g) DCCA. (h) AE2Nets. (i) MDC. (j) TMPCC. (k) Ours.

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TABLE VII
ABLATION STUDY FOR ATTENTION MODULE OF DFCPC ON THE TRENTO DATASET. BEST RESULTS ARE MARKED IN BOLD

	Trento					MUUFL					Berlin				
	ACC	**Kappa**	**NMI**	**ARI**	**Purity**	**ACC**	**Kappa**	**NMI**	**ARI**	**Purity**	**ACC**	**Kappa**	**NMI**	**ARI**	**Purity**
channel	89.84	86.43	**93.64**	94.22	90.02	58.66	47.02	50.35	48.66	**79.12**	59.06	44.04	44.49	34.95	75.77
spatial	79.12	70.55	78.26	68.25	79.12	53.68	40.01	43.56	39.68	61.41	49.42	29.84	34.35	22.95	67.83
dual-feature	**94.42**	**92.54**	91.20	**94.41**	**94.48**	**59.76**	**48.42**	**51.34**	**50.26**	65.03	**59.85**	**46.27**	**45.64**	**36.23**	**76.50**

TABLE VIII
ABLATION STUDY FOR THE LOSS FUNCTIONS OF DFCPC ON THE TRENTO DATASET. BEST RESULTS ARE MARKED IN BOLD

	Trento					MUUFL					Berlin				
	ACC	**Kappa**	**NMI**	**ARI**	**Purity**	**ACC**	**Kappa**	**NMI**	**ARI**	**Purity**	**ACC**	**Kappa**	**NMI**	**ARI**	**Purity**
Con loss only	90.04	86.71	**94.07**	**94.45**	90.09	57.90	**50.20**	**57.83**	43.91	**78.14**	48.83	37.57	40.01	24.27	75.23
Clu loss only	63.98	47.76	43.49	37.08	64.09	43.68	1.16	1.35	0.68	43.75	58.57	4.13	4.26	3.25	58.58
Joint loss	**94.42**	**92.54**	91.20	94.41	**94.48**	**59.76**	48.42	51.34	**50.26**	65.03	**59.85**	**46.27**	**45.64**	**36.23**	**76.50**

EOMSC-CA[2] is an enhanced orthogonal MVSC method that incorporates a channel attention mechanism. This mechanism improves feature representation from different viewpoints by focusing on the most relevant channels.

BMVC[3] is a bilinear model for visual classification that captures complex visual patterns by modeling the bilinear interactions between features.

FPMVS-CAG[4] integrates the channel attention and graph structure for feature-preserving MVSC. It enhances feature representation by leveraging channel attention to focus on relevant features and employing a graph structure to retain essential information.

SMVSC[5] is a self-supervised MVSC method that leverages self-supervised learning to automatically extract useful features from multiview data, enhancing subspace clustering effectiveness.

DCCA[6] is a deep learning method that learns the nonlinear correlation between two perspectives or modalities, enabling joint representation learning through a deep neural network.

AE2-Nets[7] is a self-encoder network designed for multiviewpoint learning. It employs a self-encoder structure to learn shared feature representations across multiple viewpoints, thereby enhancing the model's capability to integrate and utilize multiview data effectively.

MDC[8] is a method that integrates deep learning with multimodal data for clustering. It processes data from various modalities using deep networks and performs clustering by learning joint feature representations.

TMPCC[9] is an advanced clustering method for processing multimodal RS data. It combines the Transformer model and contrast learning strategy to extract and compare features of different modal data for more accurate clustering.

[2]https://github.com/Tracesource/EOMSC-CA
[3]https://github.com/DarrenZZhang/TPAMI19_BMVC
[4]https://github.com/wangsiwei2010/FPMVS-CAG
[5]https://github.com/Jeaninezpp/SMVSC
[6]https://github.com/Michaelvll/DeepCCA
[7]https://github.com/AkChen/AE2-Nets
[8]https://github.com/Kasra2020/MDC
[9]https://github.com/AngryCai/TMPCC

C. Classification ACC and Discussion

We present quantitative results for evaluating clustering methods using the Trento, MUUFL, and Berlin datasets, as detailed in Tables IV–VI, respectively. The proposed DFCPC method significantly outperforms others across all five metrics and datasets. Specifically, DFCPC achieves accuracies of 94.42% on the Trento dataset, 59.76% on the MUUFL dataset, and 59.85% on the Berlin dataset. These results establish DFCPC as a state-of-the-art method for clustering multimodal RS images. DFCPC has achieved notable advancements in computational efficiency, significantly reducing computation time compared with most algorithms, particularly TMPCC, across all datasets.

Table IV presents the classification results for the Trento dataset. The corresponding classification graph is shown in Fig. 3. The proposed DFCPC method demonstrates superior performance in terms of overall accuracy (OA), average accuracy (AA), and Kappa coefficients. It is important to note that the very low ACC observed for the ground type is partly due to the small sample size in this category, which constitutes only 1.59% of the total samples, resulting in a typical category imbalance issue.

Table V presents the classification results of various methods on the MUUFL dataset. The corresponding classification graph is shown in Fig. 4. The proposed DFCPC method achieves the best performance overall. Specifically, DFCPC records the highest OA at 59.76% and demonstrates the highest ACC for buildings and building shadows. This method proves to be highly competitive for classifying buildings and building shadows. Consequently, the results on the MUUFL dataset underscore the superior performance of the proposed DFCPC.

Table VI presents the classification results of various methods on the Berlin dataset. The corresponding classification graph is shown in Fig. 5. It is observed that the allotment type achieves close to zero ACC, likely due to the class's insensitivity to noise. In addition, the proposed DFCPC method demonstrates satisfactory classification ACC for each class when compared with other methods.

D. Ablation Study

1) Effectiveness of Attention Module: We used an encoder with only the channel attention module as our baseline model.

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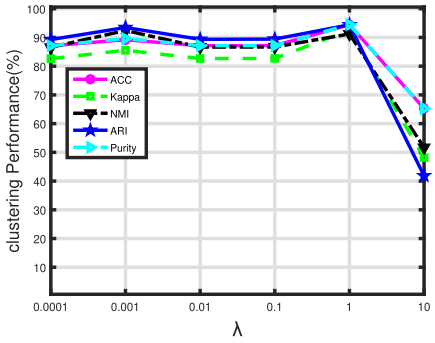


Fig. 6. Sensitivity analysis of parameters λ on the Trento dataset.

As shown in Table VII, replacing the channel attention module with the spatial attention module leads to a reduction in classification performance across the three datasets. This suggests that, despite mapping multimodal features to the same spatial size, channel eigenfunctions remain crucial for feature extraction, particularly for datasets with inherently small channel dimensions. Furthermore, using both the channel attention module and the spatial attention module in parallel significantly improves the classification performance. This enhancement occurs because fusing channel and spatial attention interactively allows the model to learn the interdependence of channel and spatial features globally, highlighting the hierarchical nature of the structure and increasing the discriminative power of the features.

2) Impact of Loss Function: Table VIII shows the comparison of different components of the proposed loss function. In order to test the effect of using only the contrastive loss (L_{ins}), we replace the cluster head with an additional k-means algorithm for cluster label prediction. The results show that the joint loss (i.e., $L_{\text{ins}} + L_{\text{clu}}$) proposed in this article significantly improves the clustering ACC on both multimodal data. For example, on the Trento dataset, the ACC using the joint loss is 4.38 and 30.44 points higher than that using only the contrastive loss (ACC $= 90.04\%$) and only the clustering loss (ACC $= 63.98\%$), respectively. This illustrates the complementary roles of contrastive loss and clustering loss in improving clustering performance. In addition, contrastive loss improves over clustering loss by 26.06 and 14.22 points on the Trento and MUUFL data, respectively, indicating its superiority in capturing clustering-friendly feature representations.

E. Parameter Sensitivity

1) Sensitivity of λ: To study the sensitivity of the equilibrium parameter, we varied λ in the $[10^{-4}, \ldots, 10^{1}]$ range of λ and analyzed its effect on the Trento dataset. The results are shown in Fig. 6. It is worth noting that when λ exceeds 1, we observe a significant decrease in performance. On the contrary, smaller values of λ tend to produce stable performance, and the best classification performance is achieved when λ is 1, suggesting that the comparison loss term and the clustering loss term reinforce each other. Therefore, in our implementation, λ is set to 1.

2) Impact of the Size of Batch and Patch: Fig. 7 demonstrates the effect of batch size and patch size on the

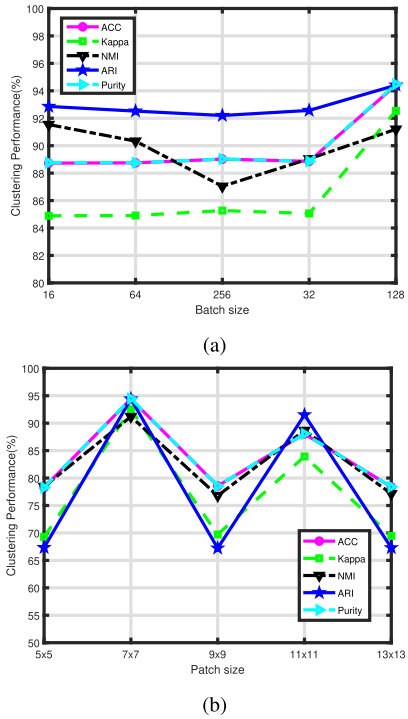


Fig. 7. Influence of (a) batch size and (b) patch size on the Trento dataset.

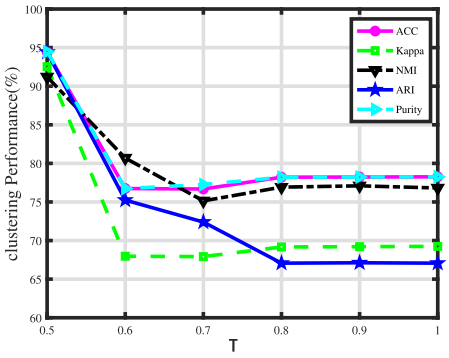


Fig. 8. Behavior of temperature coefficient on the Trento dataset.

performance of DFCPC. As shown in Fig. 7(a), clustering ACC improves with an increasing batch size. A larger batch size produces more negative samples per iteration, which effectively mitigates representation collapse, thereby enhancing the performance of DFCPC. On the other hand, Fig. 7(b) highlights the significant impact of patch size on clustering performance. Specifically, DFCPC achieves optimal performance with a patch size of 7×7. Previous studies suggest that an appropriate patch size enhances the exploration of spatial dependencies in remotely sensed data. However, a larger patch size can introduce noisy information, which explains the notable drop in clustering performance with a 13×13 patch size.

3) Behavior of Temperature Coefficient: In contrastive learning, the temperature coefficient τ is a crucial balancing hyperparameter. Fig. 8 illustrates its effect on our unsupervised multimodal learning task. When $\tau = 1$, multiple neurons are activated simultaneously, which limits the discriminative power of DFCPC in clustering tasks and results in an ACC

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of 0.78. On the other hand, when $\tau = 0.5$, the clustering performance improves significantly with an ACC of 0.94. This indicates that DFCPC can reliably assign cluster labels. Previous research has shown that lower temperatures in contrastive loss penalize hard negative samples more severely. However, excessively low temperatures can hinder the exploration of generalization between patterns, potentially leading to performance degradation. Therefore, in our implementation, τ is set to 0.5.

V. Conclusion

This article introduces a novel unsupervised multimodal clustering method for RS land cover classification: the DFCPC network. The network seeks to aggregate various land cover types into a discriminative shared space using a multimodal attention mechanism. Our method is trained end-to-end to optimize a joint loss function that combines instance comparison loss and prototype-based clustering loss. Through extensive experimental evaluations, we find that the model demonstrates high scalability, discriminative power, and adaptability in modal representation and fusion. Specifically, we conducted experiments on three widely used multimodal RS datasets, including performance comparisons and ablation studies. The results show that DFCPC achieves significant improvements across all metrics compared with existing methods.

In future work, we will explore how to further enhance the feature interaction between HSI and LiDAR data to improve the performance of deep multimodal clustering.

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